



Shenzhen Sipeeed Technology Co.,Ltd.

Sheet: /

File: Tang_Mega_60K_Console_32001C.kicad_sch

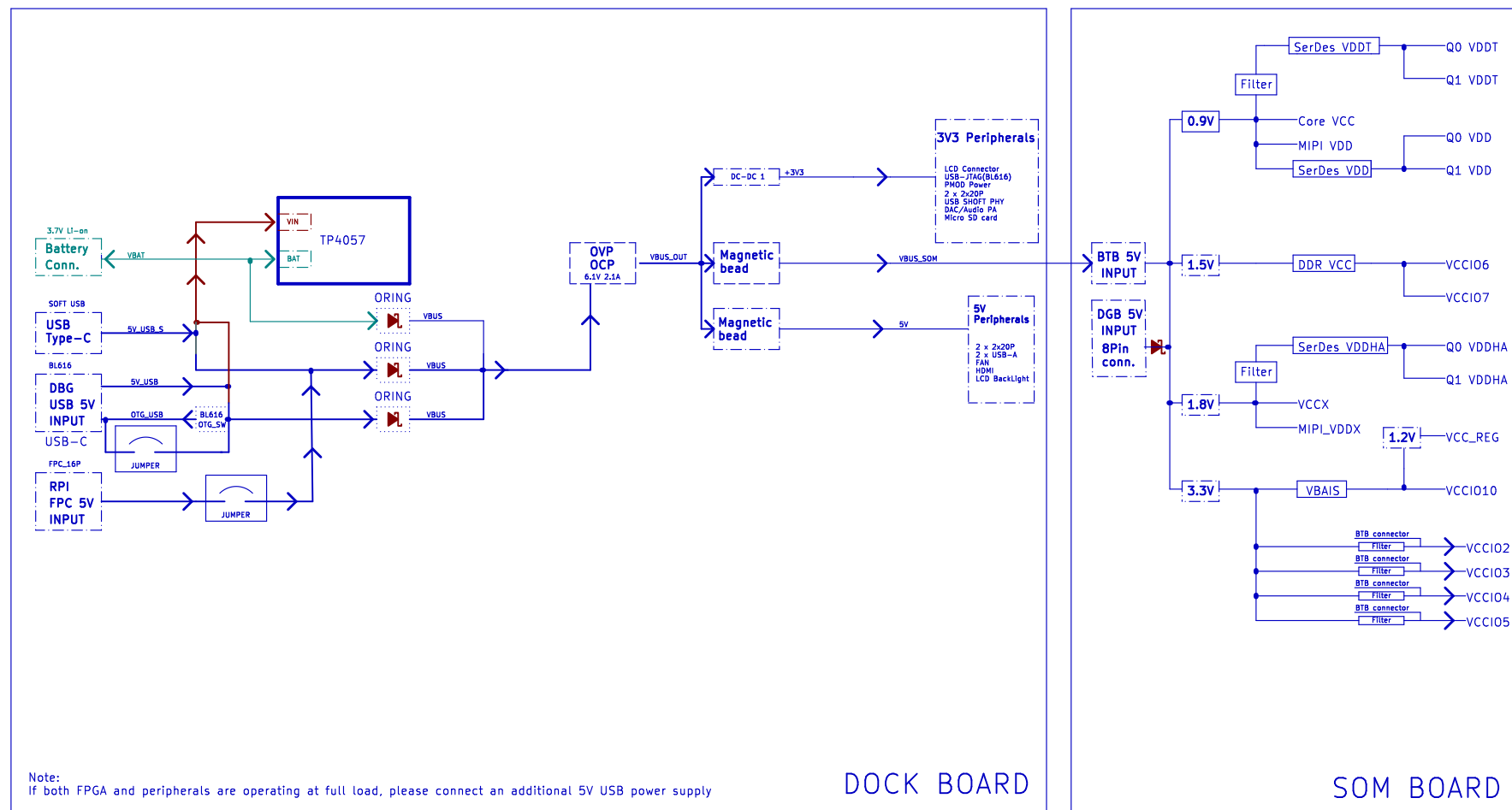
Title: Tang_60K_Console_V32001C

Size: A4	Date: 2025-04-24
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KiCad E.D.A. 9.0.1

Rev: 1.3

Id: 1/17



A Retro Ready Console Powered by FPGA

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Sheet: /POWER_TREE/

File: PWR_TREE.kicad_sch

Title: Tang_60K_Console_V32001C

Size: A4

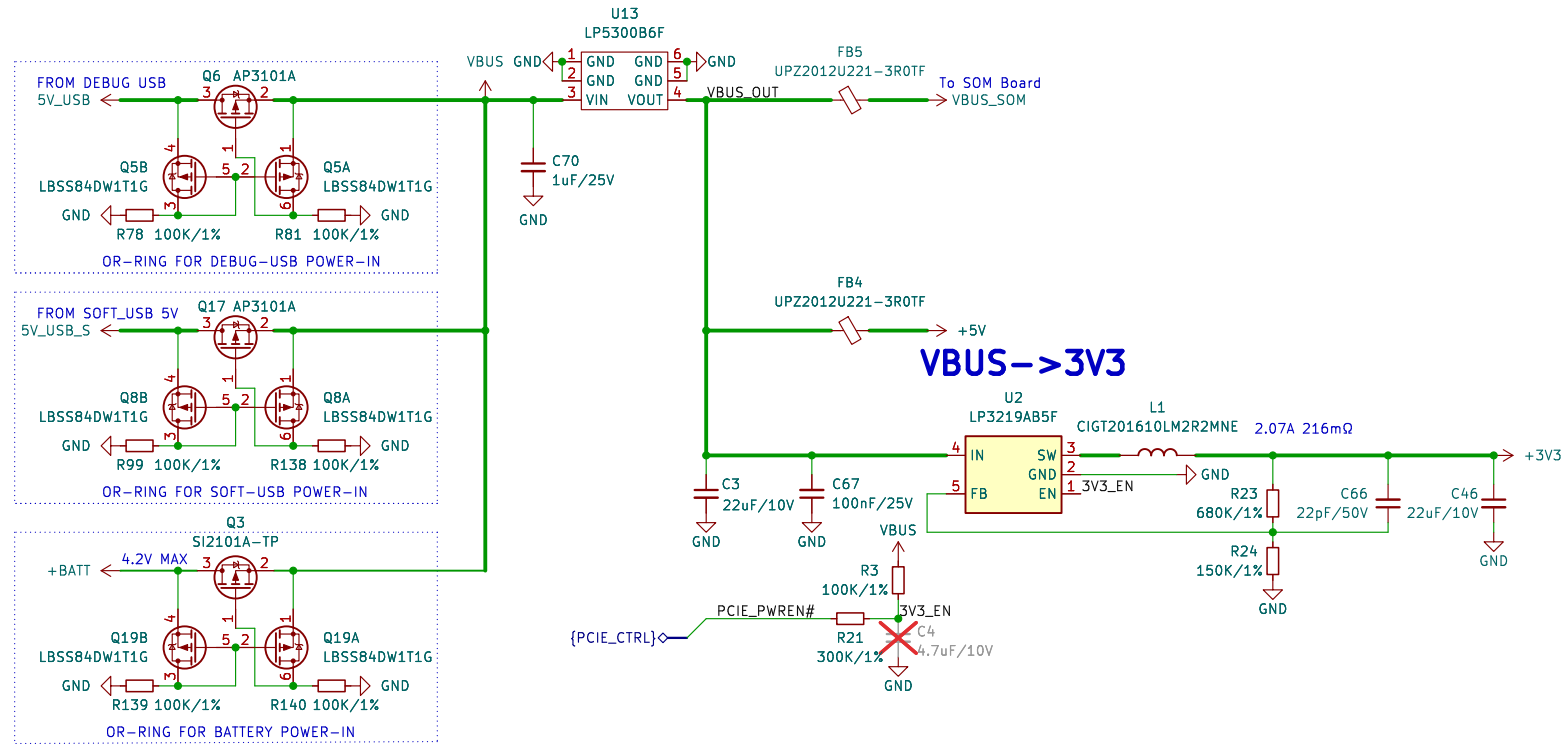
Date: 2025-04-24

Rev: 1.3

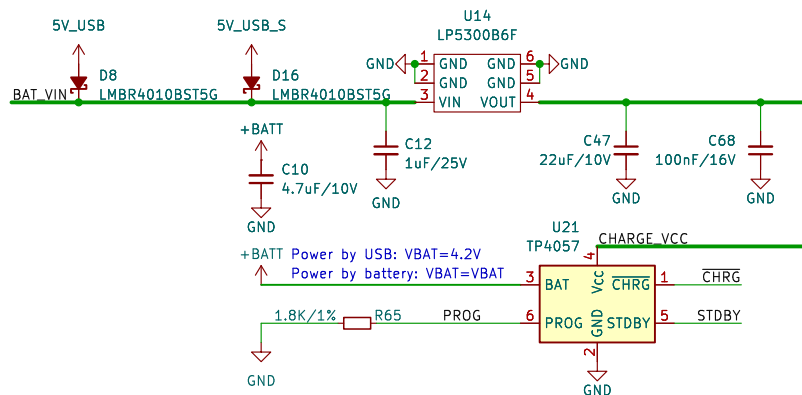
KiCad E.D.A. 9.0.1

Id: 2/17

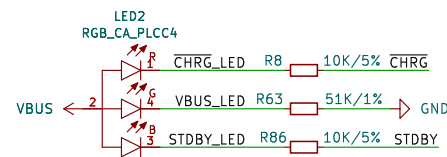
SYS POWER



Battery CHRG



VBUS_LED



Connect to
3.7V Li-on battery



A Retro Ready Console Powered by FPGA

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Sheet: /POWER/

File: PWR_MAIN.kicad_sch

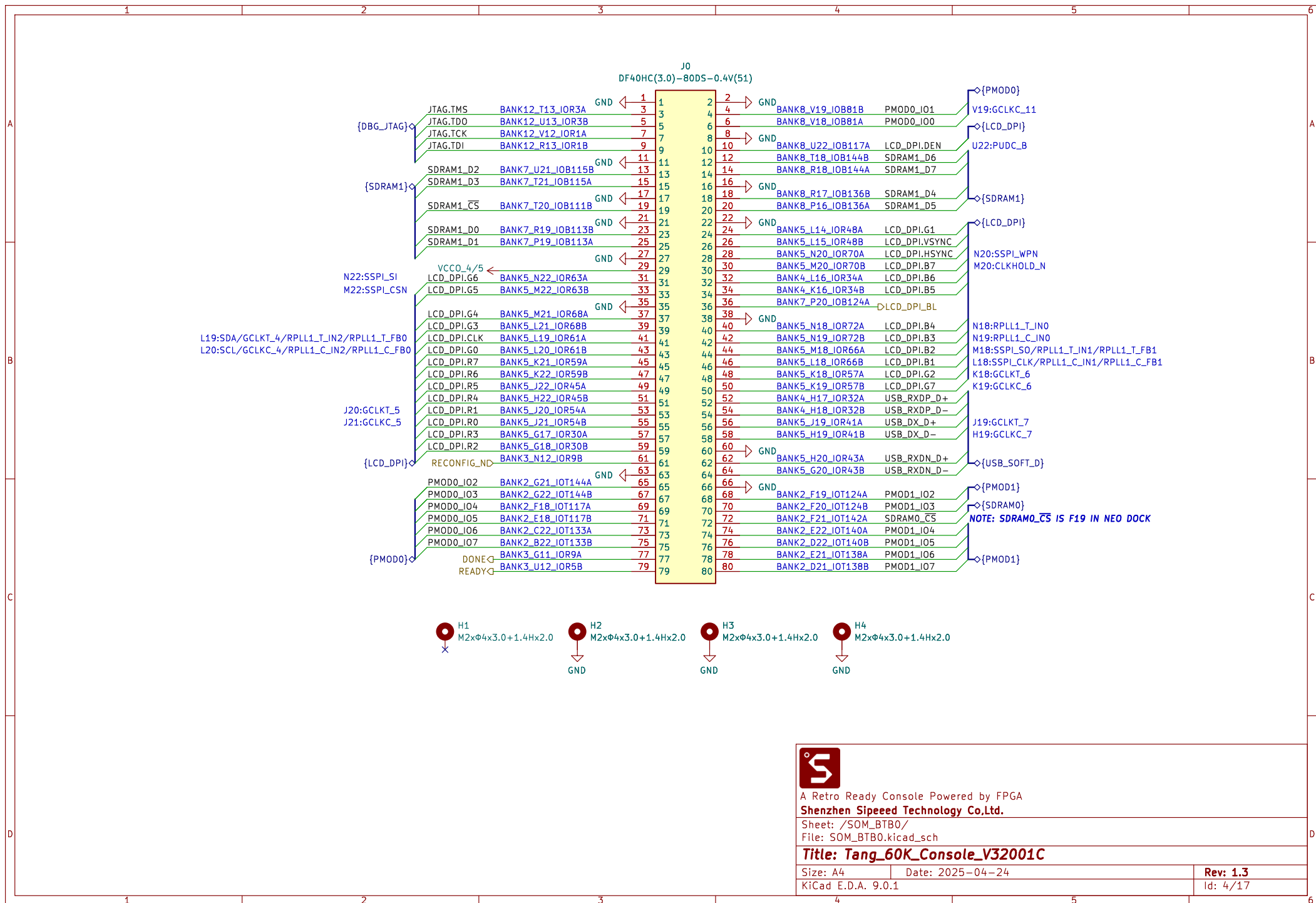
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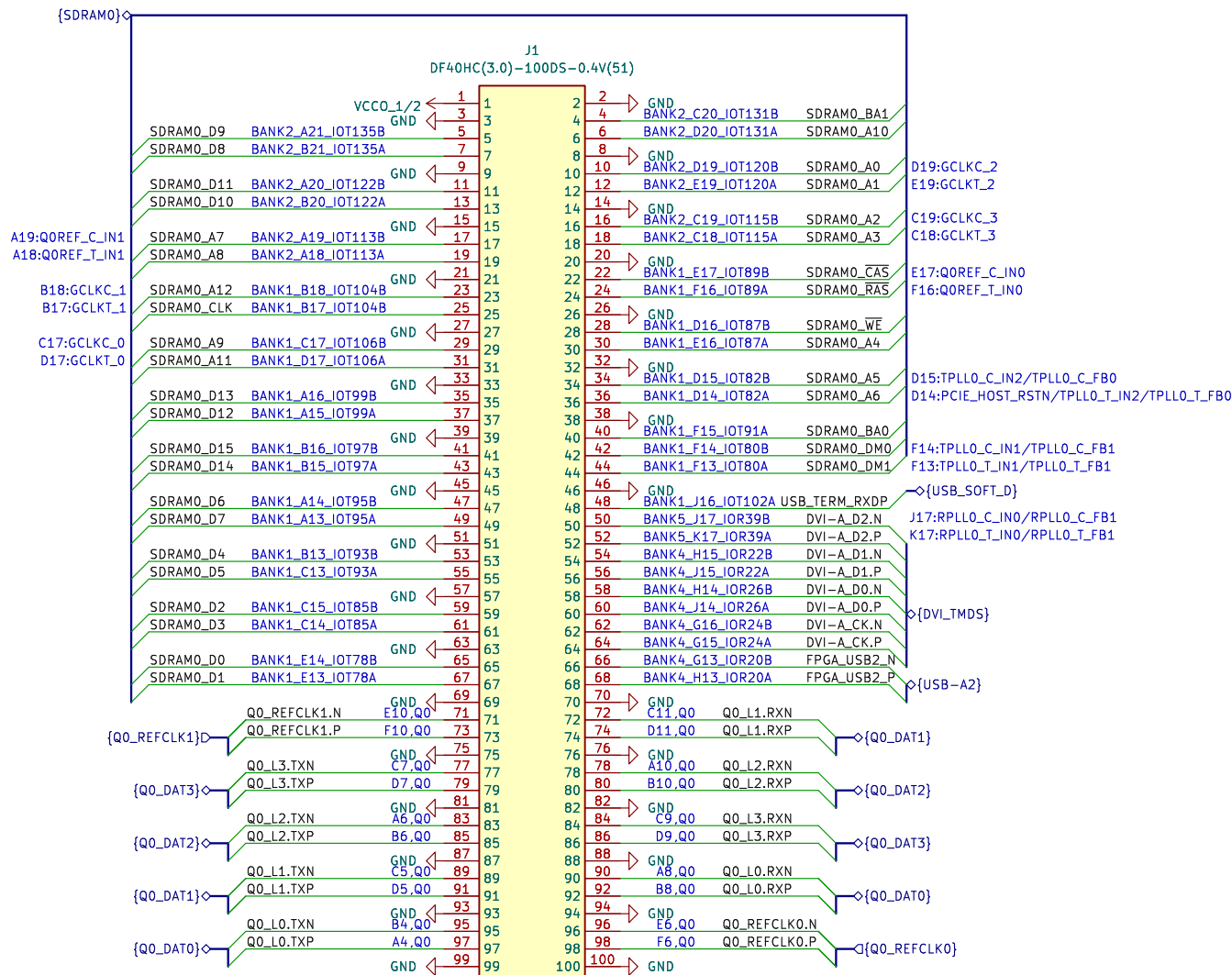
Size: A4 Date: 2025-04-24

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Id: 3/17





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Sheet: /SOM_BTBI/

File: SOM_BTBI.kicad_sch

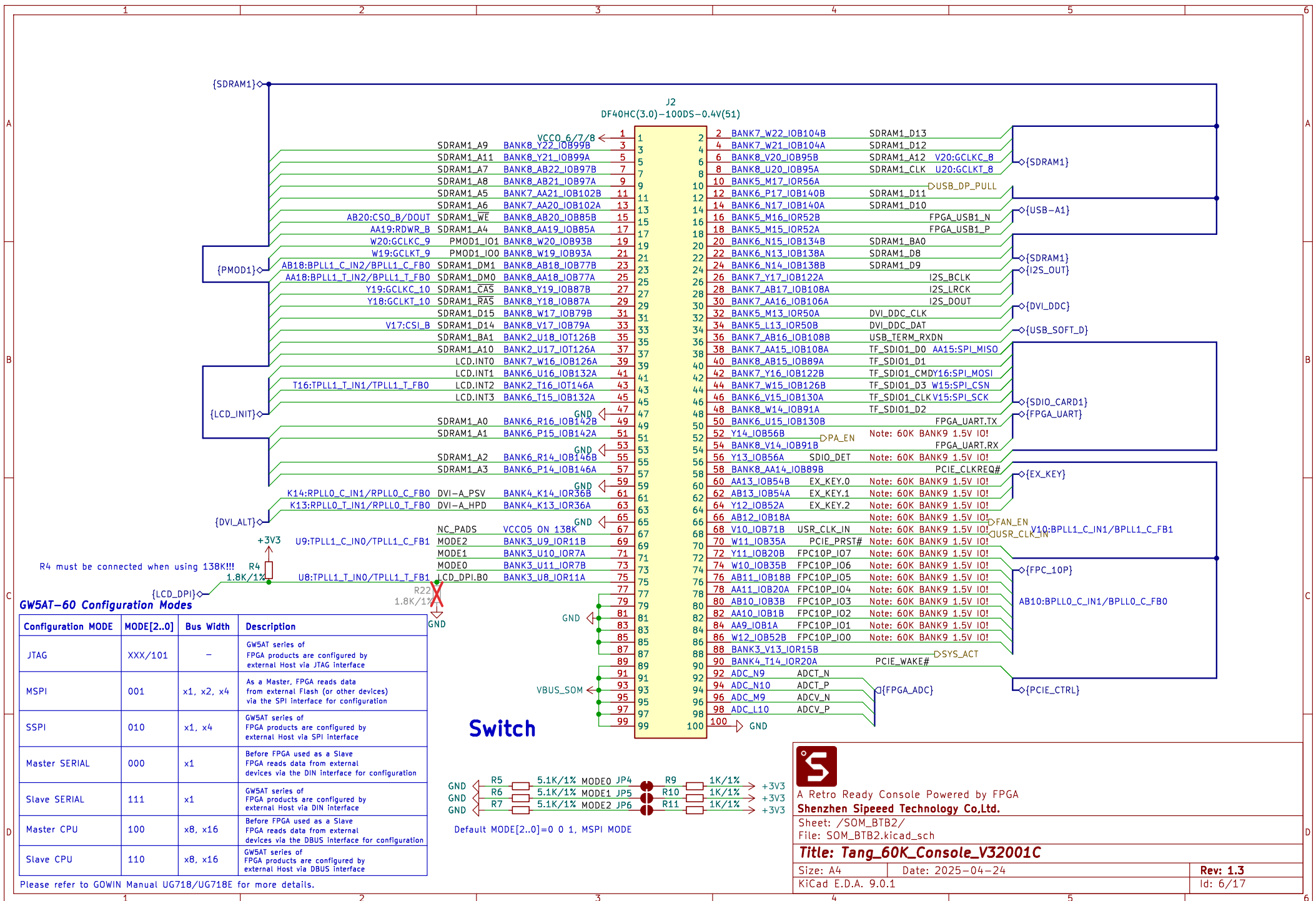
Title: Tang_60K_Console_V32001C

Size: A4 Date: 2025-04-24

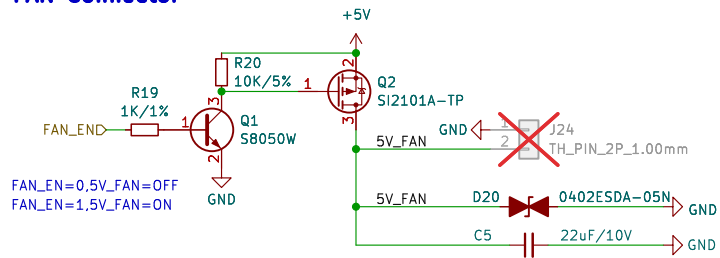
KiCad E.D.A. 9.0.1

Rev: 1.3

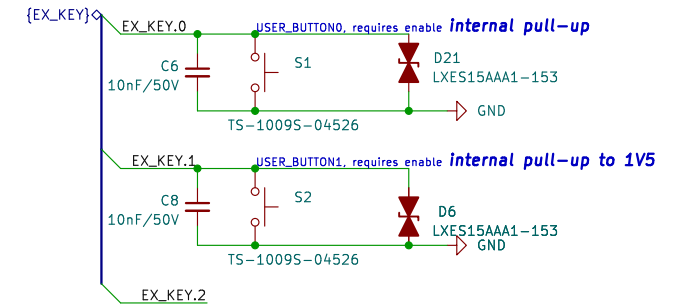
Id: 5/17



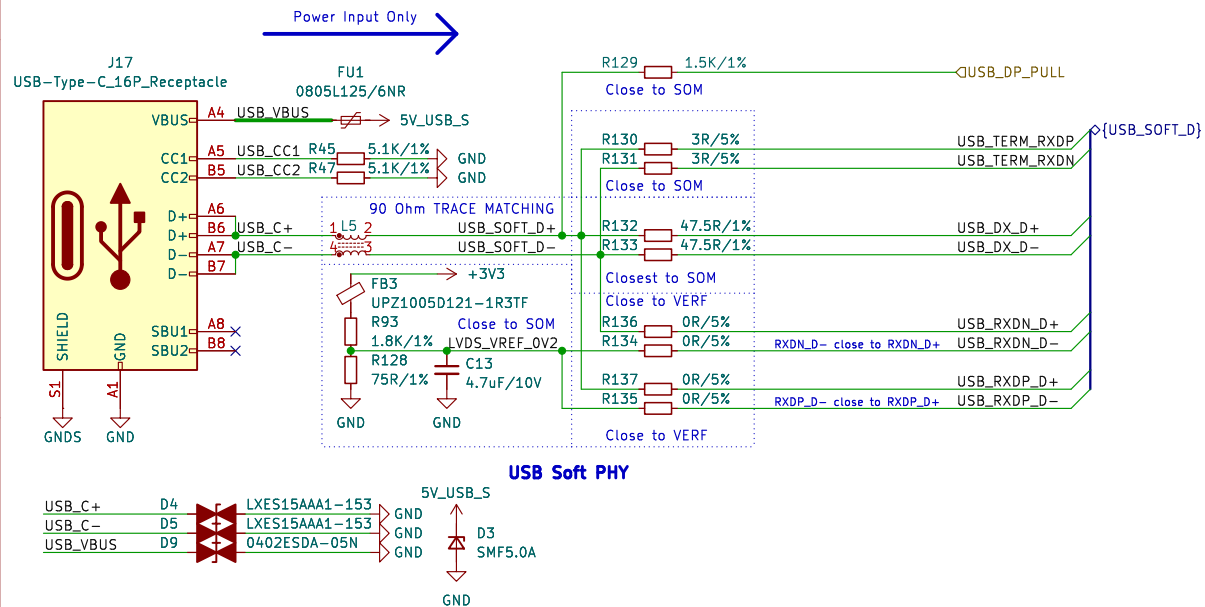
FAN Connector



BUTTON



USB Soft Deivce



A Retro Ready Console Powered by FPGA

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Sheet: /PERIPHERALS/

File: PERIPHERALS.kicad_sch

Title: Tang_60K_Console_V32001C

Size: A4	Date: 2025-04-24
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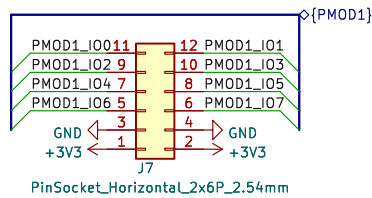
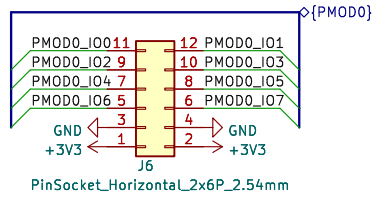
KiCad E.D.A. 9.0.1

Rev: 1.3

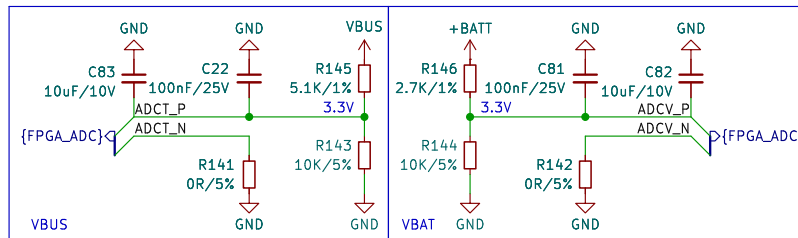
Id: 7/17

EX CONN.

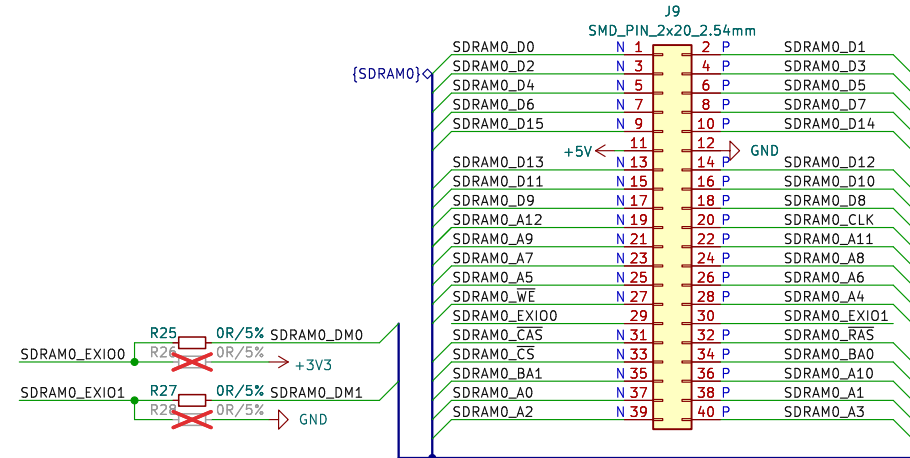
PMOD



ADC INPUT



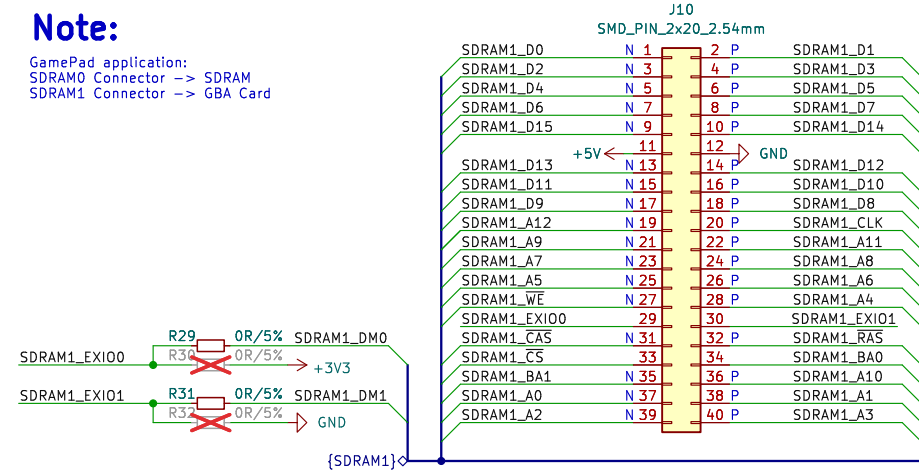
SDRAM0 CONN. Close to PCIE connector



Note:

GamePad application:
SDRAM0 Connector -> SDRAM
SDRAM1 Connector -> GBA Card

SDRAM1 CONN. Close to PMOD connector



A Retro Ready Console Powered by FPGA

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Sheet: /FPGA_EXT_CONN/

File: EX_CONN.kicad_sch

Title: Tang_60K_Console_V32001C

Size: A4

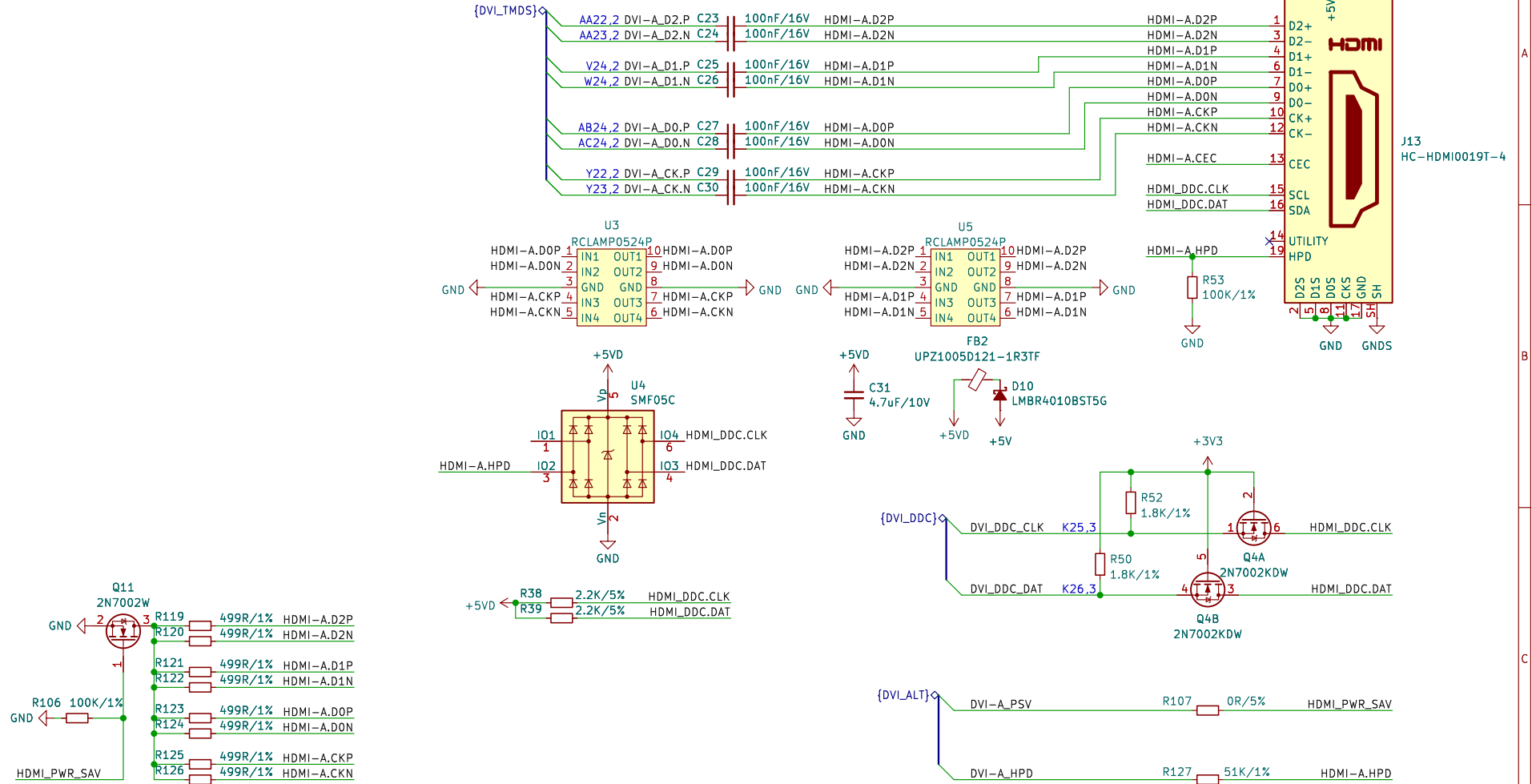
Date: 2025-04-24

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Rev: 1.3

Id: 8/17

DVI OUTPUT



A Retro Ready Console Powered by FPGA

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Sheet: /DVI_TMDS/

File: TMDS_TX.kicad_sch

Title: Tang_60K_Console_V32001C

Size: A4

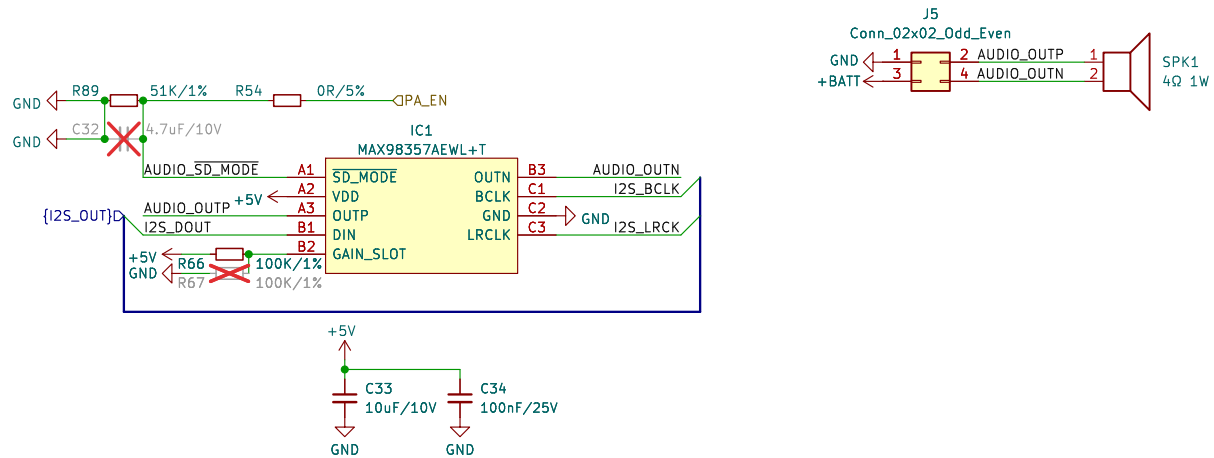
Date: 2025-04-24

KiCad E.D.A. 9.0.1

Rev: 1.3

Id: 9/17

DAC + Audio PA



A Retro Ready Console Powered by FPGA
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Sheet: /AUDIO+PA/
 File: AUDIO.kicad_sch

Title: Tang_60K_Console_V32001C

Size: A4

Date: 2025-04-24

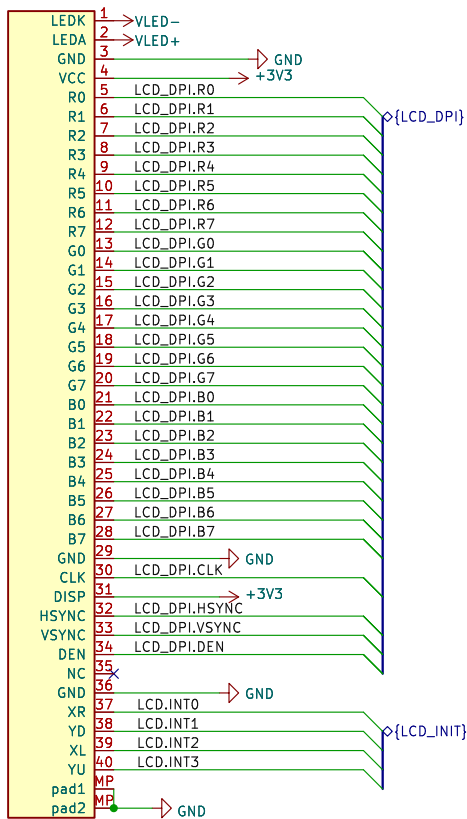
Rev: 1.3

KiCad E.D.A. 9.0.1

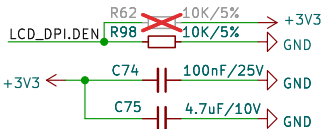
Id: 10/17

RGB LCD Display

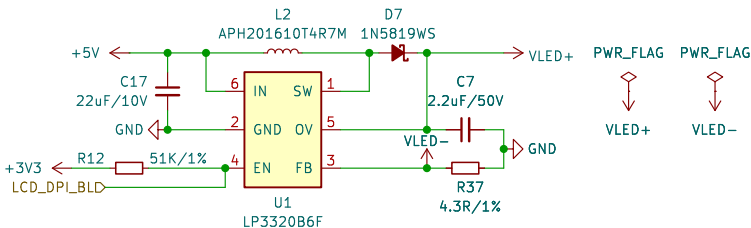
RGB LCD Display Connector



J3
FPC-40P-0.5MM-H2.0MM-rearlock



BACKLIGHT Driver



Feedback Voltage = 250mV
BackLight Current = $250\text{mV}/4.3\text{R} \approx 58.1\text{mA}$
7inch LCD QTM070TFT40-001 : IF=20*3=60mA

LCD.INT1 R13 5.1K/1% +3V3
LCD.INT0 R36 5.1K/1% +3V3



A Retro Ready Console Powered by FPGA

Shenzhen Sipeed Technology Co.,Ltd.

Sheet: /LCD_DPI_24BIT/

File: LCD_DPI.kicad_sch

Title: Tang_60K_Console_V32001C

Size: A4

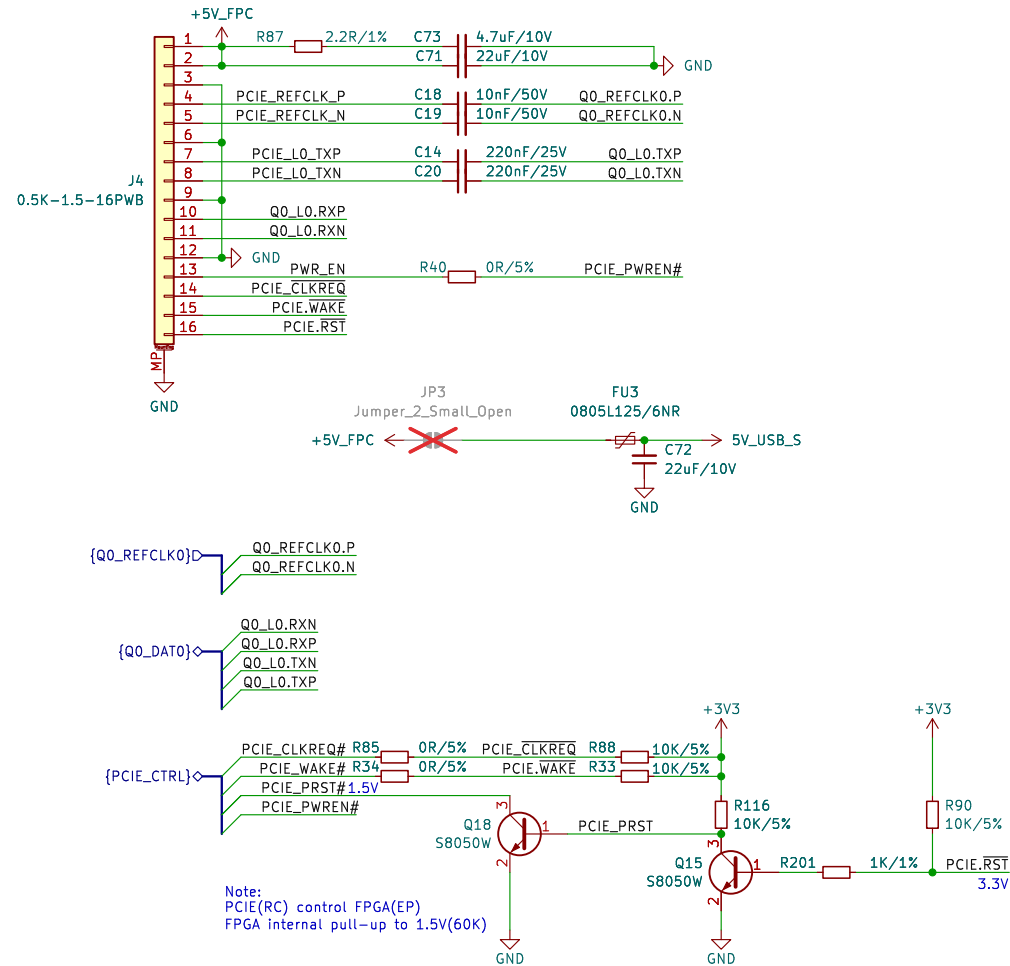
Date: 2025-04-24

Rev: 1.3

KiCad E.D.A. 9.0.1

Id: 11/17

RPI HAT+ PCIE



A Retro Ready Console Powered by FPGA

Shenzhen Sipeed Technology Co.,Ltd.

Sheet: /PCIE_GT/

File: SERDES.kicad_sch

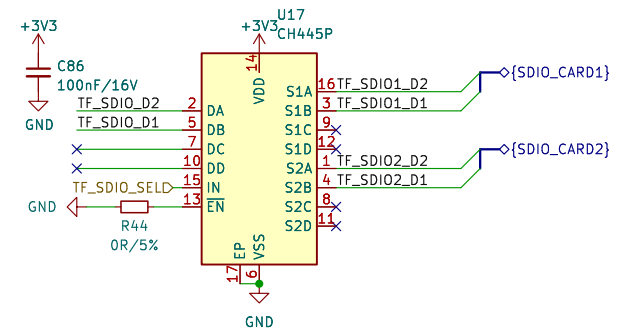
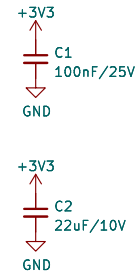
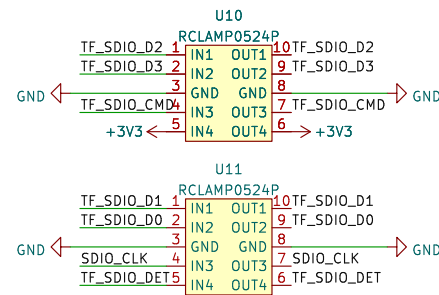
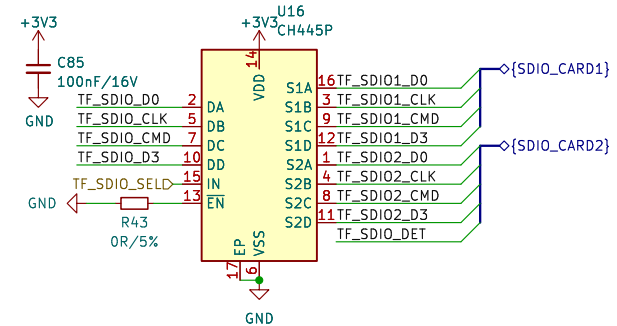
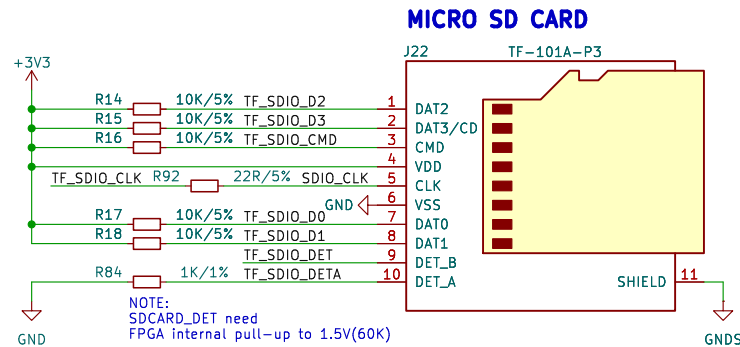
Title: Tang_60K_Console_V32001C

Size: A4 Date: 2025-04-24

KiCad E.D.A. 9.0.1

Rev: 1.3

Id: 12/17



A Retro Ready Console Powered by FPGA

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Sheet: /SDIO_CARD/

File: SDCARD.kicad_sch

Title: Tang_60K_Console_V32001C

Size: A4 Date: 2025-04-24

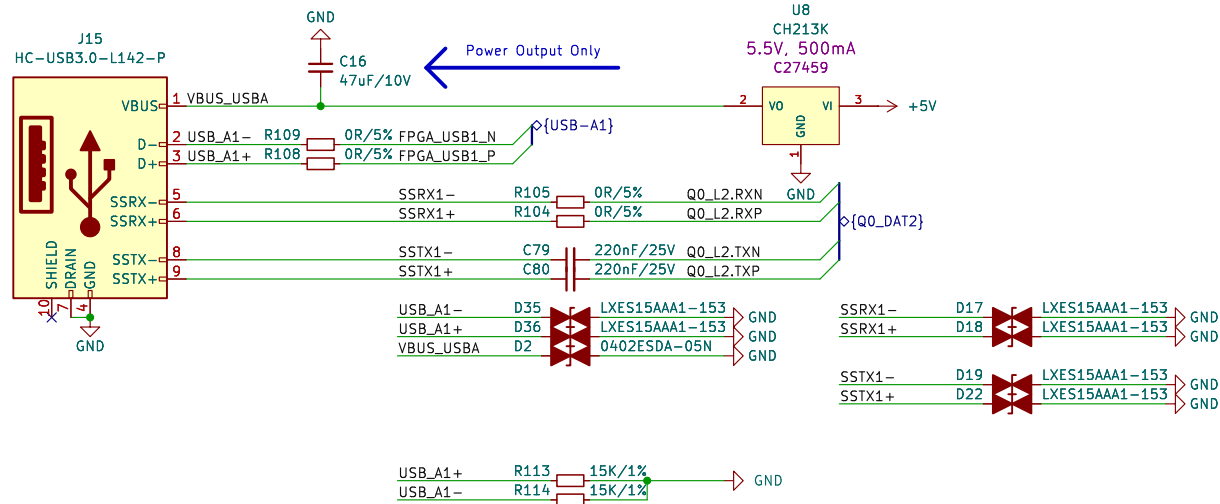
KiCad E.D.A. 9.0.1

Rev: 1.3

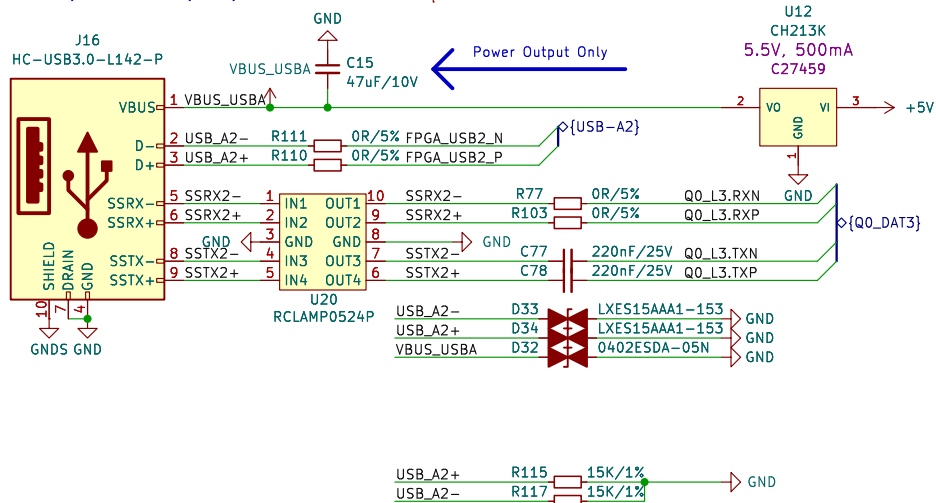
Id: 14/17

USB_Ax2

USB_A1 Full-Speed Host + Super-Speed Device



USB_A2 Full-Speed Host + Super-Speed Device



A Retro Ready Console Powered by FPGA

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Sheet: /USB3-A_GT/

File: USB3-GT.kicad_sch

Title: Tang_60K_Console_V32001C

Size: A4 Date: 2025-04-24

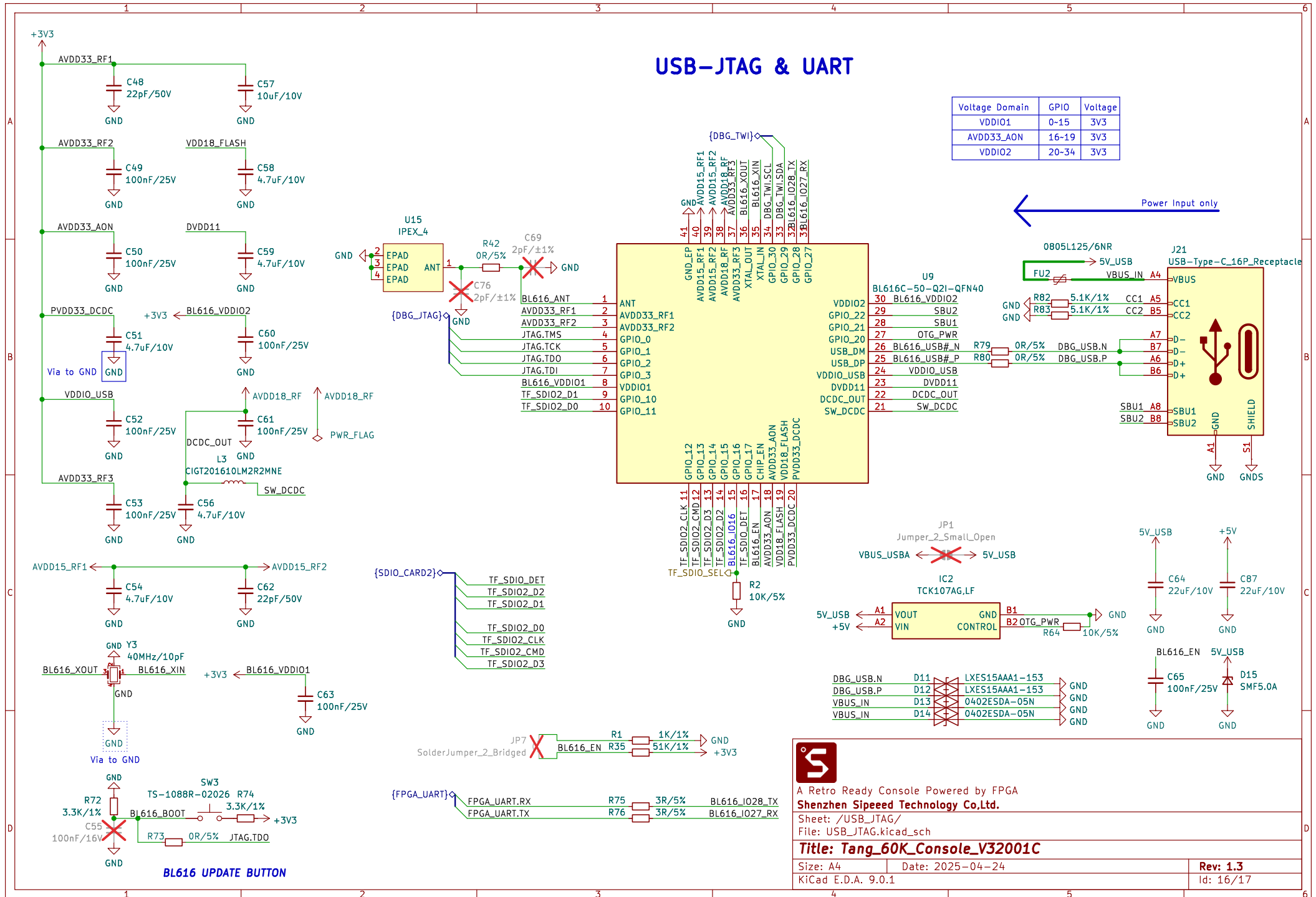
KiCad E.D.A. 9.0.1

Rev: 1.3

Id: 14/17

USB-JTAG & UART

Voltage Domain	GPIO	Voltage
VDDIO1	0-15	3V3
AVDD33_AON	16-19	3V3
VDDIO2	20-34	3V3



A Retro Ready Console Powered by FPGA

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Sheet: /USB_JTAG/

File: USB_JTAG.kicad_sch

Title: Tang_60K_Console_V32001C

Size: A4 Date: 2025-04-24

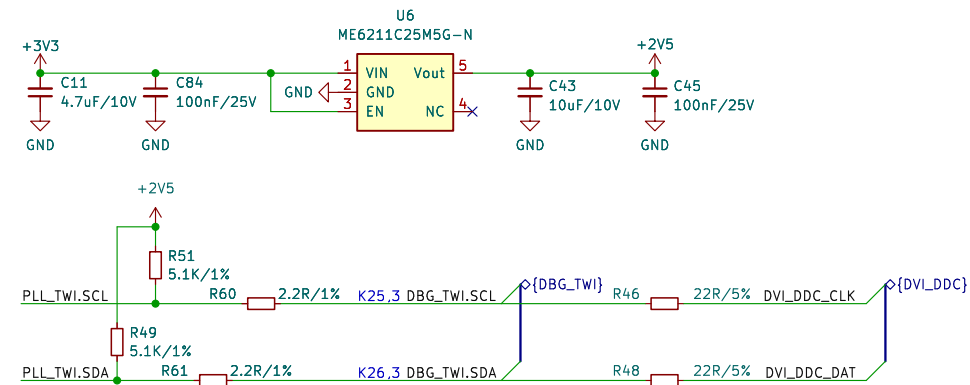
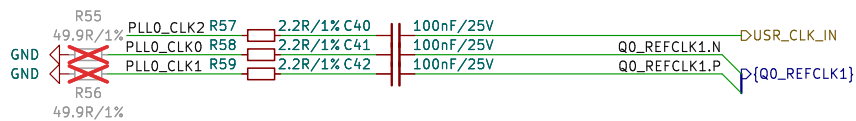
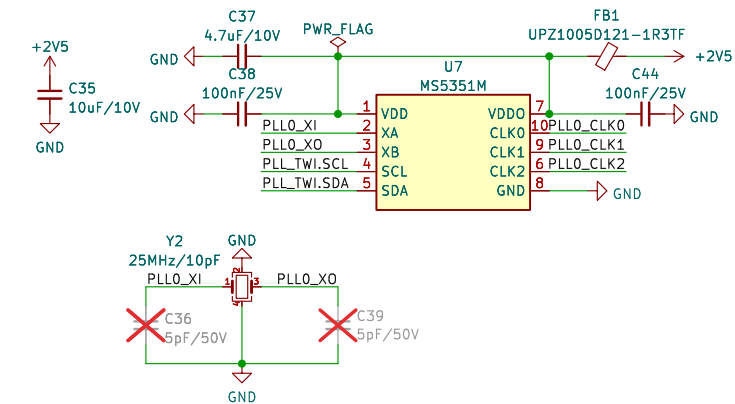
KiCad E.D.A. 9.0.1

Rev: 1.3

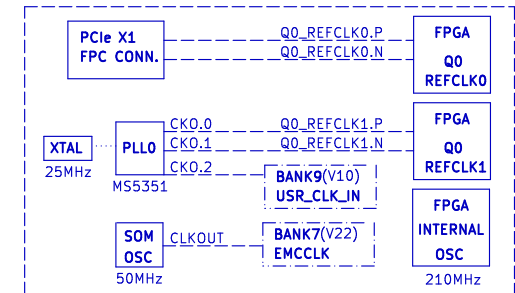
Id: 16/17

CLOCK

EXT. PLL



COLCK TREE



FPGA internal clock generator OSC.
The user can change the value of the option "Frequency Divider" to generate a different output clock frequency, which can be one of 64 values: 1/2, 1/3, 1/4, 1/6, 1/8, 1/10, or 1/126 of the oscillator frequency.



A Retro Ready Console Powered by FPGA
Shenzhen Sipeed Technology Co.,Ltd.

Sheet: /SYS_CLOCK/
File: CLOCK.kicad_sch

Title: Tang_60K_Console_V32001C

Size: A4	Date: 2025-04-24
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KiCad E.D.A. 9.0.1

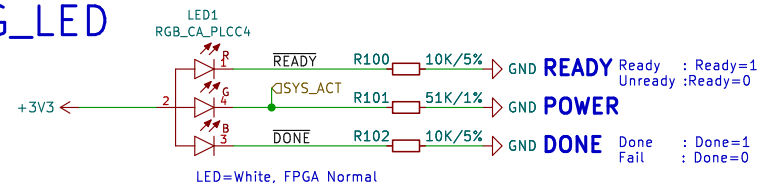
Rev: 1.3

Id: 18/17

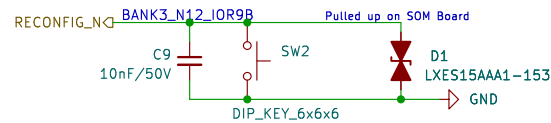
Inverter Circuit



CFG_LED



RECFG_KEY



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Shenzhen Sippeed Technology Co.,Ltd.

Sheet: /FPAG_CONFIG/
File: FPGA_CFG.kicad_sch

Title: Tang_60K_Console_V32001C

Size: A4 Date: 2025-04-24

KiCad E.D.A. 9.0.1

Rev: 1.3

Id: 19/17